

Observational Indistinguishability and Integrity Blind Regions in Hybrid Quantum-Classical Workflows

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Abstract—Integrity auditing asks which evidence lets an auditor distinguish a consequential workflow intervention from its baseline. We model hybrid quantum-classical workflows through observational indistinguishability under declared views and reference profiles, separating reference provenance, granularity, and decision rule. The constructive result is claim-relative minimum evidence granularity: a trusted aggregate anchor for the same batch suffices to certify integrity of a reported conclusion, whereas item-identity integrity requires item-aligned provenance; some root outside the adversary’s control is unavoidable. Across eight frozen environments from CICIDS2017, UNSW-NB15, and ToN-IoT, 3,600 label interventions confirm the structural boundaries. The batch sensors execute statistically thresholded aggregate comparisons with protected clean same-item-set benchmark oracles, not authenticated deployment evidence. A conformal family rule has finite-sample validity under exchangeability; observed false-action rates of 0.048–0.058 in the non-exchangeable design are descriptive. On the equal-weight grid, feature-bearing batch policies serve 4,365–4,496 of 7,008 materially altered audit results. A cluster-preserving adaptive stress test of the fragile fingerprint reduces matched-strength detection in the feature and feature-plus-prediction regimes from 0.96–1.00 to 0.01–0.66 while retaining 83–91% of materiality; its served fraction varies sharply with strength. A bounded 165-cell ideal-statevector and finite-shot-emulation gate maps the same distinctions onto circuit, kernel, and estimation boundaries without making a hardware claim.

Index Terms—hybrid quantum-classical systems, integrity auditing, observability, quantum kernels, workflow integrity

1 INTRODUCTION

HYBRID quantum-classical software distributes one reported result across heterogeneous computational and evidential boundaries. Classical records are acquired, cleaned, projected and encoded; a circuit and execution stack produce quantum evidence; a learner turns a kernel into a decision; and an evaluation layer aligns that decision with reference outcomes and reports a metric. A plausible final number may therefore coexist with a failure in the data, circuit, kernel, prediction, label or reporting path. Security analyses of

hybrid systems consequently call for lifecycle-aware controls rather than trust in an isolated algorithm [1], [2].

Robustness benchmarking collapses this chain into a performance change. That endpoint cannot identify what changed or whether the auditor could have seen it. Consider a hybrid security service that scores each batch with a fixed predictor, then joins the predictions to a ground-truth feed from another subsystem (analyst verdicts, incident tickets or a delayed label store). The predictor is intact, but the join or store is stale, corrupted or rewritten, so balanced accuracy moves although the model has not. Feature and prediction monitors cannot see the event because their inputs are unchanged. Label counts expose an unbalanced corruption but not a balanced exchange of class identities. A signature authenticates the label file, not its item-level alignment with the predictions, so a correctly signed stale or misaligned file still passes.

Calling the event stealthy therefore requires a qualifier: stealthy to which auditor, holding which evidence and trusting which reference? The constructive result is not to store another copy of every label. A trusted aggregate anchor for the same batch certifies that the honest labels give the reported conclusion; only item-identity integrity requires item alignment. If one party controls both the label store and its checking reference, no local verifier can distinguish a substituted honest pair from the original (Proposition 6). The paper identifies the minimum external root and its granularity for the protected claim; it does not remove the need for that root. The scenario is a reading aid, not an incidence estimate.

Prior art already establishes observation-relative detectability [3]–[5], authenticates evaluation artifacts through VAMP [6], constructs drifts that evade their detectors [7], and verifies quantum-stage contracts and behavioural evidence on hardware [8], [9]. We claim none of these ingredients. Our result derives claim-relative minimum evidence granularity within a workflow-wide information-set lattice that includes the evaluation-label path. The experiments validate its structural predictions, measure calibrated policy consequences, instantiate residual adaptive evasion against a declared fingerprint, and map the distinctions onto a bounded quantum branch.

The contribution is one package, led by the claim-relative granularity result:

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- an observation model of hybrid workflows with primitive and derived artifacts, in which auditability is a property of (intervention class, information set, trusted references), with monotonicity, closure and materiality results, minimal counterexamples, and the quantum branch as an instance of the same lattice with semantic, estimated and observed kernels kept apart;
- an explicit adversary and failure model for every executed intervention, with the roots it cannot write, including an executed adaptive attacker who preserves the fingerprint that batch-level sensors rely on;
- multi-dataset, fixed-OOD validation with verified deduplication, cluster-level uncertainty and preregistered null calibration;
- decision-level calibration by a conformal family rule whose finite-sample level holds under exchangeability, and an offline `allow/hold/block` analysis of 24,000 policy rows derived from the frozen intervention grid; inferential independence resides at the prespecified environment/split clusters rather than at row level; and
- a bounded quantum-path validation and a four-contract prototype that execute the composed decision locally.

No constituent is claimed as individually sufficient novelty: the propositions are finite-batch statements, the family rule is a standard conformal construction, contracts, hash chains and semantics-preserving tests have clear prior art, and the secondary quantum-versus-classical impact profile (supplement) is neither a quantum-advantage result nor a vulnerability ordering. Physical hardware, calibrated backend noise, scheduling, provider-side security, multi-tenant interference, deployed runtime services and operational Fleet Management are outside the claim.

2 RELATED WORK AND POSITIONING

2.1 Stealth, Detectability and Observability in Secure Systems

Observation-relative detectability is established: cyber-physical attacks can be unobservable, residual-stealthy or confined to an estimator’s range space [3]–[5]; later work quantifies the detectability–impact trade-off and the residuals a monitor can form [10]–[12]. Hinder *et al.* construct drifts designed to evade drift detectors [7]. Gate A is a concrete cluster-preserving instance against our declared batch fingerprint; it measures materiality, policy consequences and evidence regimes without claiming priority for adaptive drift evasion. Simplex supplies the canonical decision/fallback pattern [13], [14], runtime verification checks executions [15], and integrity checkers compare digests with trusted baselines [16]. Our claim is not generic information-relative detectability, but blind regions and calibrated decisions for workflow views whose protected boundaries include evaluation ground truth.

2.2 Monitoring, Labels and Evaluation Integrity

Monitoring work treats shifted priors, selective labels and proxy sufficiency under delayed ground truth [17]–[20].

Benchmark-quality work treats label error, evaluation pitfalls, spatial/temporal sampling bias and measurement blindness [21]–[24]. These lines primarily model labels as shifted, delayed, noisy, erroneous or measurement-limited. VAMP [6], by contrast, explicitly treats evaluation artifacts as authenticated assets that may be poisoned or substituted. Our distinction is therefore not that evaluation data can be attacked, but that auditability and trusted-reference granularity are derived relative to the protected claim and the auditor’s view.

VAMP protects named datasets, software, models and evaluation sets through authenticated provenance, providing a concrete external root of the kind required by Proposition 6. Its focus is artifact authentication; our complementary question is which view and granularity suffice for conclusion, aggregate or item-identity integrity, and what remains indistinguishable with weaker evidence. A VAMP-style manifest can instantiate the references assumed here.

In-toto, SLSA and Sigstore bind supply-chain steps, build provenance, identity and transparency [25]–[27]. Remote attestation supplies fresh evidence to an appraiser [28]; RATS names Attesters, Verifiers, Evidence, Reference Values and appraisal policy [29]. These mechanisms establish provenance or appraisal, not whether aggregate or item-aligned evidence suffices for a claim. Provider/runtime authentication is the operational scope of Paper 2.5.

2.3 Calibration of Multi-Sensor Decisions

The family rule of Section III-D is a conformal p -value [30], [31]: it combines per-sensor ranks by Tippett’s minimum- p statistic and calibrates the maximum as in Westfall–Young [32], [33]. The standard exchangeability argument remains valid with ties counted against rejection [30], [34]. Conformal anomaly detection, shared-calibration p -values and dependence-aware Max-Rank provide the closest statistical context [35]–[37]. We claim no novelty for the rule: the leave-one-out family score calibrates an information-regime decision under its stated premise and exposes the executed design’s premise violation. Beyond-exchangeability conformal methods exist [38], but this paper does not implement one. The asymmetric comparison rule had no such guarantee (Proposition 5(c)).

2.4 Hybrid and Quantum Software Integrity

Hybrid-system analyses decompose control, compilation, execution and measurement surfaces and cover malicious compilation, hardware faults, cloud exposure and side channels [1], [2], [39]. Quantum Vulnerability Factor scores fault sensitivity, while Quantum Leak demonstrates cloud timing leakage [40], [41]. QCIVET combines stage contracts, hash-chained traces, behavioural subtyping, calibrated observable tests and QPU validation [8]. QML-PipeGuard uses a structured observable family and calibrated drift tolerance to detect channel substitution on IBM hardware [9]; multi-level work evaluates circuit-integrity metrics [42].

QProv records provider-independent provenance across circuit, compilation, execution and hardware context [43]; recent cross-provider work extends this line toward unified quantum-job provenance [44]. Differential, metamorphic and equivalence-modulo-input tests address the oracle problem

[45]–[47]; compiler verification, program logics and runtime assertions supply other formal/runtime mechanisms [48]–[52]. We claim no priority for these mechanisms. QCIVET and QML-PipeGuard reason directly about quantum-stage contracts and behavioural evidence, including calibrated observables, drift and hardware validation; QProv and cross-provider work address provenance. They are stronger here on QPU/hardware and runtime/provider evidence. Our orthogonal contribution is a workflow-wide, claim-relative information-set lattice that protects the evaluation-label path and distinguishes conclusion, aggregate and item-identity integrity. Its quantum evidence is simulator-only, its hash chain unsigned, and its policy offline.

2.5 Relation to Companion Work

Three companion studies by the authors are disclosed to delimit the claim. *Sharp target-domain certificates* [53] bounds the advantage of a fixed quantum-kernel candidate under distribution shift (shared data sources and fidelity kernels; an identification interval, not the observability of interventions). *Conditional validity of quantum event classifiers* [54], the closest relative, conditions fail-closed verdicts on a declared information set and exhibits exact sensor blindness to weight-only nuisances in collider data; here the conditioning is applied to the integrity of the evidence chain under interventions, with trusted references, calibrated policy and executable response. *Candidate comparability before promotion* [55] decides whether a challenger should replace a deployed detector after a drift alarm (shared benchmarks and label-free monitors). The present Paper 1.5 concerns workflow integrity, observational indistinguishability, evidence granularity and integrity policy. Planned Paper 2.5 concerns longitudinal operational QPU and context assurance. Data sources, fidelity-kernel implementations and some monitors are shared and are not presented as independent contributions; none of the propositions, interventions, tables or experiments reported here appears in the three companion manuscripts.

3 OBSERVATION MODEL AND BLIND REGIONS

The complete statement with proofs is in the supplement. The results are elementary by design: their role is to make exact which evidence separates which intervention class, so that the gates test statements rather than intuitions.

3.1 States, Interventions and Materiality

A workflow state separates stored artifacts from those the workflow derives. The primitive artifacts of one evaluation run are

$$s = (X, P, C, E, \xi, g, f, y) \in \mathcal{S}, \quad (1)$$

acquired features X , preprocessing P , circuit or feature map C , execution context E , estimation randomness ξ , post-processing map g , fitted predictor f and evaluation labels y . The derived artifacts are recomputed from them: the representation $\tilde{X} = P(X)$; the semantic kernel $K_{\text{sem}} = \Phi(C)$ induced by the circuit on a fixed probe set; the finite-shot estimate $\hat{K} = \text{est}(K_{\text{sem}}, E, \xi)$; the observed kernel $K_{\text{obs}} = g(\hat{K})$ delivered downstream after estimation, transmission and post-processing; the predictions $\hat{y} = f(\tilde{X}, K_{\text{obs}})$,

indexed by item identity $i = 1, \dots, n$; and the result $R = r(\hat{y}, y)$, balanced accuracy of $\hat{y}$ against y . An intervention a overwrites a declared set of artifacts, primitive or derived, keeps every non-descendant fixed and recomputes the descendants through the workflow with ξ retained; the baseline is s_0 and $s_a = a(s_0)$. Label-only T_y overwrites y item-wise, so only R is recomputed (its prior-preserving subclass T_y^π also preserves the class histogram); feature-side mechanisms overwrite $\tilde{X}$ or X and recompute $\hat{y}$ and R ; circuit-side interventions overwrite C and recompute K_{sem} and everything below it; estimation variation overwrites ξ or E with K_{sem} fixed; post-processing overwrites g or K_{obs} with K_{sem} and $\hat{K}$ fixed. “Fixes everything else” thus means every non-descendant. The signed conclusion change is

$$\Delta_R(a) = R(s_0) - R(s_a). \quad (2)$$

An intervention is *material at level* τ if $|\Delta_R(a)| \geq \tau$. The limit $\tau \rightarrow 0^+$ is a structural-sensitivity endpoint—whether the reported conclusion changes at all—not an operational risk threshold or an assertion that every epsilon has the same consequence. The fixed $\tau = 0.02$ and 0.05 analyses show larger effect sizes; service-level risk is outside this study. Earlier versions used the positive part $\max(\Delta_R, 0)$; the signed endpoint counts an apparent improvement caused by an integrity failure as a conclusion change.

3.2 Views, Refinement and Trusted References

An information set $\mathcal{I}$ is a view map $V_{\mathcal{I}} : \mathcal{S} \rightarrow \mathcal{O}_{\mathcal{I}}$; the auditor observes $V_{\mathcal{I}}(s)$ and nothing else. The regimes studied are $\mathcal{I}_X$ (the multiset of rows of $\tilde{X}$), $\mathcal{I}_{XF}$ (the multiset of pairs $(\tilde{x}_i, \hat{y}_i)$), $\mathcal{I}_{Y_m}$ (the class-count vector of y), $\mathcal{I}_{XFY}$ (the multiset of triples $(\tilde{x}_i, \hat{y}_i, y_i)$) and $\mathcal{I}_Q$ (circuit, kernel and execution evidence). $\mathcal{I} \sqsubseteq \mathcal{I}'$ (refinement) iff $V_{\mathcal{I}} = \pi \circ V_{\mathcal{I}'}$ for some π ; hence $\mathcal{I}_X \sqsubseteq \mathcal{I}_{XF} \sqsubseteq \mathcal{I}_{XFY}$ and $\mathcal{I}_{Y_m} \sqsubseteq \mathcal{I}_{XFY}$, while $\mathcal{I}_{Y_m}$ is incomparable with $\mathcal{I}_X$ and $\mathcal{I}_Q$ is a separate branch. The join $\mathcal{I} \vee \mathcal{W}$ is the pair of views.

A *reference profile* separates three dimensions: provenance (historical, benchmark-protected, or deployment-authenticated), granularity (aggregate or item-aligned), and decision rule (statistical or exact). A reference is *trusted* if no intervention in the class under study can alter it. The A–C labels are shorthand, not a one-dimensional trust scale. Class A is a *statistically thresholded aggregate comparison*; its reference may be historical or, as executed here, a protected clean same-item-set oracle used without item pairing. The oracle is outside the benchmark attack API, but no deployed authentication mechanism is demonstrated. Class B is an exact invariant against a trusted aggregate same-batch reference, such as $M(s_0)$, $\text{hist}(y_0)$ or $R(s_0)$. Class C is an exact invariant against a trusted item-aligned same-batch reference such as $(y_{0,i})_i$. Thus statistical versus exact and aggregate versus item-aligned are not synonyms for untrusted versus trusted. For a label-path intervention, *item-identity integrity* holds if $y_a = y_0$ item-wise, *aggregate integrity* if $M(s_a) = M(s_0)$ and *conclusion integrity* if $R(s_a) = R(s_0)$; each implies the next and no converse holds. $\mathcal{I}^*$ denotes a regime augmented with trusted same-batch references ($\mathcal{I}_{XFY}^*$ holds the aggregate confusion profile and item-aligned labels and predictions; feature coverage remains statistical). A label available in $\mathcal{I}_{XFY}$ is not thereby trusted:

if the label store is the asset under attack, the auditor sees y_a , not y_0 .

3.3 Equivalence, Sensors and Blind Regions

Definition 1. $s \sim_{\mathcal{I}} s'$ iff $V_{\mathcal{I}}(s) = V_{\mathcal{I}}(s')$. A sensor admissible in $\mathcal{I}$ is a measurable $S : \mathcal{O}_{\mathcal{I}} \rightarrow \mathbb{R}^k$, written $S(s) = S(V_{\mathcal{I}}(s))$.

Lemma 1 (structural blindness). If $s_a \sim_{\mathcal{I}} s_0$ then $S(s_a) = S(s_0)$ for every sensor admissible in $\mathcal{I}$ (path-wise for a seeded randomized sensor; in distribution for a fresh seed).

Definition 2. For a class $\mathcal{A}$, the structural blind region of $\mathcal{I}$ is $B_{\mathcal{I}}(\mathcal{A}) = \{a \in \mathcal{A} : a(s_0) \sim_{\mathcal{I}} s_0\}$ and the sensor blind region of a family $\mathcal{S}$ is $B_{\mathcal{I},\mathcal{S}}(\mathcal{A}) = \{a : S(a(s_0)) = S(s_0) \forall S \in \mathcal{S}\}$. By Lemma 1, $B_{\mathcal{I}} \subseteq B_{\mathcal{I},\mathcal{S}}$, with equality iff $\mathcal{S}$ is *baseline-separating* on the orbit of $\mathcal{A}$ (a different value at $a(s_0)$ than at s_0 whenever the views differ); detection needs baseline separation, identification would need pairwise separation, which is not claimed. Sensor blindness thus decomposes into information-level blindness and sensor insufficiency; the adaptive attacker of Section VI-D exploits the second. The structural and sensor auditability gaps at level τ are

$$G_{\mathcal{I}}(\tau) = \{a : |\Delta_R(a)| \geq \tau, a \in B_{\mathcal{I}}\}, \quad G_{\mathcal{I},\mathcal{S}}(\tau) \supseteq G_{\mathcal{I}}(\tau). \quad (3)$$

Proposition 1 (monotonicity). If $\mathcal{I} \sqsubseteq \mathcal{I}'$ then $B_{\mathcal{I}'}(\mathcal{A}) \subseteq B_{\mathcal{I}}(\mathcal{A})$ and $G_{\mathcal{I}'}(\tau) \subseteq G_{\mathcal{I}}(\tau)$ for every τ .

Proof: $V_{\mathcal{I}'}(s_a) = V_{\mathcal{I}'}(s_0)$ implies $V_{\mathcal{I}}(s_a) = \pi(V_{\mathcal{I}'}(s_a)) = \pi(V_{\mathcal{I}'}(s_0)) = V_{\mathcal{I}}(s_0)$. $\square$

Corollary 1 (label-path boundaries). Let f be fixed and deterministic. (a) $T_y \subseteq B_{\mathcal{I}_{XF}} \subseteq B_{\mathcal{I}_X}$: every sensor admissible in $\mathcal{I}_X$ or $\mathcal{I}_{XF}$ is invariant under a label-only change. (b) $T_y^\pi \subseteq B_{\mathcal{I}_{Ym}}$: every sensor admissible in $\mathcal{I}_{Ym}$ is invariant under a prior-preserving change.

Proposition 2 (closure). $B_{\mathcal{I} \vee \mathcal{W}}(\mathcal{A}) = B_{\mathcal{I}}(\mathcal{A}) \cap B_{\mathcal{W}}(\mathcal{A})$; a class $\mathcal{A} \subseteq B_{\mathcal{I}}$ becomes completely separable after adding $\mathcal{W}$ iff $V_{\mathcal{W}}(a(s_0)) \neq V_{\mathcal{W}}(s_0)$ for every $a \in \mathcal{A}$.

Corollary 2 (label-path closure). No view factoring through $(\tilde{X}, f(\tilde{X}), \text{hist}(y))$ separates any $a \in T_y^\pi$. The multiset $\mathcal{I}_{XFY}$ separates it unless labels move only among items with identical $(\tilde{x}, \hat{y})$; an item-aligned view separates every non-identity relabeling.

Proposition 3 (materiality forces aggregate separability). Let $R = g(M)$ be a function of the confusion matrix $M(s)$, as balanced accuracy is. If $a \in T_y$ and $\Delta_R(a) \neq 0$ then $M(s_a) \neq M(s_0)$; hence a is separable in the confusion view and in $\mathcal{I}_{XFY}$, and $G_{\mathcal{I}_{XFY}}(\tau) \cap T_y = \emptyset$ for every $\tau > 0$.

Proof: Contrapositive of $M(s_a) = M(s_0) \Rightarrow R(s_a) = R(s_0)$; the confusion view is a coarsening of the multiset of triples. $\square$

The proposition holds for every metric that is a function of the binarised confusion matrix; for a ranking metric such as ROC-AUC the same argument holds with the multiset of (score, label) pairs as the aggregate (supplement). No AUC experiment is run.

3.4 Three Auditor Classes and Decision-Level Calibration

A *reference-anchored* auditor holds a trusted same-batch reference ρ (class B or C) and computes $D(s) = d(V_{\mathcal{J}}(s), \rho)$

with a metric d ; then $D(s_0) = 0$ exactly, the rule “fire iff $D > 0$ ” has zero false-action probability, and separability in $\mathcal{J}$ is sufficient for detection. A *batch-level statistical* auditor computes aggregate scores and thresholds them against a clean calibration population (class A). Its scoring reference can be historical or same-item-set; either way, separability need not yield power beyond clean-score variability. A *post-hoc certifier* recomputes a view from inputs it trusts (the evaluation contract recomputes R from $(\hat{y}, y)$) and needs trusted inputs rather than a stored value.

Proposition 4. (i) For any auditor whose decision is a function of $V_{\mathcal{I}}(s)$, $a \in B_{\mathcal{I}}$ implies $\text{fire}(s_a) = \text{fire}(s_0)$ on every batch, so the calibrated detection rate equals the false-alarm rate. (ii) A reference-anchored auditor detects every $a \notin B_{\mathcal{J}}$, deterministically.

Corollary 3 (which reference certifies which integrity). Let $a \in T_y$ with f fixed. (a) With a trusted aggregate reference $M_0 = M(s_0)$ of the same batch, “fire iff $M(s_a) \neq M_0$ ” detects every violation of aggregate integrity and, by Proposition 3, every material a : conclusion and aggregate integrity need only level B. (b) Every reference that factors through M , $\text{hist}(y)$ or R is invariant under relabelings that permute labels among items with equal predictions (C1); certifying item identity needs level C, and an item-aligned y_0 detects every non-identity relabeling.

Proposition 5 (union versus conformal family calibration). (a) If m sensors fire under the null with probabilities $p_j \leq \alpha$, then $\max_j p_j \leq \Pr_0[\bigcup_j E_j] \leq \min(1, \sum_j p_j) \leq \min(1, m\alpha)$, the bounds being attained by nested and disjoint events. (b) Let $v_1, \dots, v_n$ be the sensor vectors of n calibration draws and v_{n+1} that of the audited batch; give every member j of the augmented set the leave-one-out family score $\tilde{U}_j = \max_s \#\{i \neq j : v_{s,i} < v_{s,j}\}/n$, let $p = (1 + \#\{i \leq n : \tilde{U}_i \geq \tilde{U}_{n+1}\})/(n+1)$, and fire iff $p \leq \alpha$ (ties against firing). At most $\lfloor \alpha(n+1) \rfloor$ members of any augmented set would fire if audited; hence, if the $n+1$ draws are exchangeable, $\Pr_0[p \leq \alpha] \leq \lfloor \alpha(n+1) \rfloor / (n+1) \leq \alpha$ with no continuity or tie assumption [30], [32], [33]; $10/201 = 0.0498$ for $n = 200$, $\alpha = 0.05$. For one sensor without ties the rule is the per-sensor rule of Gate N. The level is marginal over the joint draw of calibration set and audited batch; nothing is claimed when the premise fails. (c) The asymmetric rule of the earlier version, which scored calibration draws against the other $n-1$ draws only, has no such guarantee: with three sensors and cyclic orderings over five vectors it fires with probability 0.60 at $\alpha = 0.2$ under exchangeability (supplement), and on the frozen draws it exceeds α under exchangeable re-splits where the conformal rule does not (Section VI-B).

Proposition 6 (no local authentication without an uncontrolled root). Let a local verifier be any map $\text{acc}(V_{\mathcal{I}}(s), \rho)$ of the observed view and a reference bundle, with honest references $\rho_H(s) = V_{\mathcal{J}}(s)$. If it accepts every honest pair and the class can produce $(V_{\mathcal{I}}(s'), \rho_H(s'))$ for some honest $s' \not\sim_{\mathcal{I}} s_0$ (joint control), the substitution is accepted as an honest run of s' : no function of that input establishes authenticity relative to s_0 , and if $R(s') \neq R(s_0)$ a material change is served. A bundle component $V_{\mathcal{K}}(s_0)$ the class cannot write rejects every substitution with $V_{\mathcal{K}}(s') \neq V_{\mathcal{K}}(s_0)$ (Proposition 4(ii)): authenticity requires a root outside the declared class. The result concerns authenticity, not consistency; by Proposition 3

a material substitution does change the joint view.

3.5 Counterexamples

Eight minimal constructions separate the events a robustness number conflates; the supplement tabulates them with the frozen expansion observations that realise each. C1 swaps the labels of two items with equal predictions: M , the histogram and R are invariant while item identity changes (808 of 3,600 label rows), an integrity violation with no conclusion impact, visible only item-wise against a trusted reference. C2–C4 separate marginal invariance, confusion-matrix invariance and conclusion invariance, C5–C6 do the same on the feature side, C7 shows that the clipped “harm” endpoint of the earlier evidence hid 1,534 conclusion changes (433 on the label path), and C8 is the empirical face of Proposition 3 (2,617 of 2,617 material label rows change M). Of the 3,600 label rows, a trusted aggregate reference of the same batch detects 2,792 (every material one, Corollary 3a) and the remaining 808 are C1 relabelings that only an item-aligned reference exposes (Corollary 3b). A ninth construction, the cluster-preserving drift of Section VI-D, is not a structural blind region but sensor insufficiency against an adaptive attacker, measured rather than proved.

3.6 The Quantum Branch as an Instance of the Lattice

Let $\mathcal{C}$ be the circuit representations (canonical OpenQASM 3 with bound parameters), $\Phi : \mathcal{C} \rightarrow \mathcal{K}$ the semantic map to the ideal fidelity kernel on a fixed probe set, $K_{\text{sem}} = \Phi(C)$, $\hat{K}$ and K_{obs} the estimated and observed kernels of Section III-A, $h(C)$ the provenance hash, $A(\cdot)$ the algebraic invariants and $f_K(\hat{X})$ the decision computed from a kernel. Three intervention classes are kept apart: (a) circuit-side, overwriting C ; (b) estimation variation, with C and K_{sem} fixed and $\hat{K}$ changing; (c) post-processing or kernel substitution, with C , K_{sem} and $\hat{K}$ fixed and K_{obs} changing. Each inclusion below names its class; none is claimed across classes.

Proposition 7 (quantum lattice). Assume h collision-free on the circuits under study. (i) [class (a)] $[C] = \Phi^{-1}(\Phi(C))$ is the semantic class of C ; approved transpilation and common-unitary rewrites map C into $[C]$ without fixing $h(C)$, so provenance refines the semantic view, $B_h \subseteq B_{K_{\text{sem}}}$, strictly whenever the class contains an approved rewrite: separability in provenance is not harm, and the auditor needs the approved class, implemented as semantic equality of probe kernels against the trusted $K_{\text{sem},0} = \Phi(C_0)$ (level B). (ii) [class (c)] A and f_K coarsen the observed kernel, so $B_{K_{\text{obs}}} \subseteq B_A$ and $B_{K_{\text{obs}}} \subseteq B_f$; a PSD-preserving substitution K' with $A(K') = A(K_{\text{obs},0})$ lies in $B_A \setminus B_{K_{\text{obs}}}$ and, since $h(C)$ and the semantic probe of the unchanged circuit see nothing, is closed only by an anchored comparison of K_{obs} itself: the trusted semantic probe and the trusted observed-kernel reference are different anchors. A kernel change that crosses no decision boundary lies in $B_f \setminus B_{K_{\text{obs}}}$. (iii) [class (b)] For an honest finite-shot estimate, exact equality $\hat{K} = K_{\text{sem},0}$ has false-alarm probability $1 - \Pr[\hat{K} = K_{\text{sem},0} \mid \text{honest}]$, which depends on the discrete support of the estimator and may be large or equal to one but is not universally one (fidelity 1/2 at two shots gives 1/2). Exact equality is therefore not an appropriate acceptance criterion; $d(\hat{K}, K_{\text{sem},0})$ must

be treated as a level-A statistic with a null from repeated honest estimation, to which Proposition 5(b) applies with one sensor when the repeated and audited estimates are exchangeable, and where no such null is calibrated no statistical integrity claim is made. The proposition adds no quantum mechanics: it names the two features the classical branches lack, an approved equivalence class coarser than provenance and an estimation step that turns an exact anchor into a statistical one. Hash chaining of the audit envelope is a reference-anchored comparison over the record itself; its root is not assumed uncontrolled, so Proposition 6 applies.

4 ADVERSARY AND FAILURE MODEL

Table 1 assigns every executed intervention to one of four classes with the roots it cannot write and the regime that identifies it (complete per-class record in the artifact). Evaluation-label classes stand for corruption of the ground-truth store or label join; feature-side classes for corruption or drift of the acquisition path feeding both branches; the cluster-preserving class for an attacker who knows the batch-level fingerprint; circuit and kernel classes for corruption of the circuit/parameter store, the compiler output or the post-estimation kernel; shot emulation for legitimate estimation uncertainty. Equal weighting is a stress profile, not a threat distribution; no prevalence is estimated.

The evidence trusts the local runtime and host, the training data and fitted model, the reference input and preprocessing declaration, the reference circuit and probe kernels, SHA-256 collision resistance, the item-level label reference only where $\mathcal{I}_{XY}^*$ is declared, and the code that builds and verifies the envelope. Compromise of the operating system, runtime or verifier; provider identity, forged job or calibration metadata and provider attestation; malicious schedulers, physical mapping and unapproved passes on hardware; device-calibrated noise, crosstalk, multi-tenant interference and QPU faults; side channels and circuit confidentiality; denial of service; and the operational Fleet Management System are outside this study.

5 METHODOLOGY

5.1 Data, Pipelines and Interventions

The frozen first gate uses a balanced binary subset of CICIDS2017 [56]: 3,000 rows, 77 numeric columns and 128/128 capped training/evaluation samples. The expansion adds eight fixed environments E1–E8 (supplement): five from CICIDS2017 (a 256/256 scale gate and four temporal source-target pairs), two from UNSW-NB15 [57] (a balanced ID subset and a temporal pair) and one from ToN-IoT [58] (a balanced ID subset), all at 128/128 unless stated and projected to dimensions 8, 10 and 12. CICIDS2017 therefore carries six of the nine environments including Gate 1; the environments are not a probability sample of deployments. Train-defined preprocessing is applied without evaluation labels: TruncatedSVD per branch, training-fitted standardization for the RBF support-vector classifier (SVC) and frozen angular scaling to $[0, 2\pi]$ for the fidelity-kernel branches (ZZ, PauliXYZ and a Z/PauliXZ-equivalent profile, one repetition). Gate 1 uses the Qiskit Machine Learning

TABLE 1

Adversary and failure model of the executed interventions. Roots: assets the class cannot write. Identifying regime: the least evidence that separates the class in the frozen design under the conformal family rule; "exact" means an item-aligned trusted reference; (a)–(c) are the intervention classes of Proposition 7.

Mechanism and capability (boundary)	Class	Roots it cannot write	Identifying regime	Supported claim
Random label flip: rewrite r of y at random (evaluation/report)	Integrity corruption	X, P, C, K, f ; item-aligned y^* if declared	exact only; $\mathcal{I}_{Y_m}/\mathcal{I}_{XFY}$ batch 0.00–0.01	invisible to $\mathcal{I}_X, \mathcal{I}_{XF}$ (Cor. 1a); undetected at budget by marginal or batch joint evidence at these rates
Prior-preserving label flip: pairwise swaps preserving class counts (evaluation/report)	Adaptive attacker	as above	exact only; $\mathcal{I}_{XFY}$ batch 0.00–0.01	invisible to $\mathcal{I}_X, \mathcal{I}_{XF}, \mathcal{I}_{Y_m}$ (Cor. 1); every material instance changes M (Prop. 3)
Confusion-preserving relabeling: swaps among equal predictions (realised, not executed)	Adaptive attacker	as above	item-aligned reference only	item-identity violation without aggregate or conclusion impact (C1; 808 rows)
Feature sign flip: negate p of $\tilde{X}$ entries (acquisition/preprocessing)	Integrity corruption	training data, fitted model, reference features	$\mathcal{I}_X$ 0.09–0.72; $\mathcal{I}_{XF}$ 0.13–0.84; exact 0.79–0.91	partially separable; strength-dependent residual
Mean shift, scaling drift: shift or scale all features (acquisition/preprocessing)	Fault robustness	as above	$\mathcal{I}_X$ 0.99–1.00 by cluster fingerprint	detected in this design by a fragile fingerprint
Feature dropout: impute p of entries (preprocessing)	Fault robustness	as above	batch 0.00–0.03; exact 0.49–0.76	separable in $\mathcal{I}_{XF}$ yet statistically undetectable with a level-A reference
Cluster-preserving mean shift and scaling: same mechanisms on the 32% of entries outside tight clusters (executed, Gate A)	Adaptive attacker	as above	$\mathcal{I}_X$ 0.01–0.56, $\mathcal{I}_{XF}$ 0.06–0.66 at matched strengths; exact for every material row	evades the fingerprint while keeping 83–91% of the conclusion changes; served by P2 in 28–39% of material rows
Circuit-parameter mutation; repetition change: alter C before execution (design; (a))	Adversarial attack	reference circuit hash, probe kernel $K_{\text{sem},0}$	circuit hash and semantic probe 15/15	identified by anchored provenance and semantics; outputs miss sub-decision changes
Benign transpilation; common-unitary rewrite: compiler pass ((a))	Approved equivalence	–	hash changes 15/15, K_{sem} unchanged	separability in $h(C)$ is not harm (Prop. 7(ii))
Asymmetric, diagonal-eroded kernel edit: edit K_{obs} after estimation ((c))	Integrity corruption	reference kernel, algebraic invariants	algebra 15/15	detected by structural invariants of K_{obs}
PSD-preserving kernel substitution: valid-looking K_{obs} ((c))	Adaptive attacker	reference kernel	algebra 0/15; circuit hash 0/15; anchored K_{obs} comparison 15/15	sensor-level blindness closed only by a reference on the observed kernel (Prop. 7(ii))
Binomial shot emulation: finite-shot estimate (execution; (b))	Fault robustness	–	repeated estimation 15/15 in these cells	held for adjudication; exact equality is not the criterion (Prop. 7(iii)); not hardware evidence
Post-hoc envelope edit: rewrite stored contracts (record)	Adversarial attack	none assumed	hash chain	tamper-evident only; no identity or non-repudiation

reference evaluator [59]; the expansion uses a tagged exact-statevector evaluator validated against it at 10^{-10} on unit matrices and by an end-to-end replay (supplement); neither is shot-noise or QPU evidence. The intervention suite has six mechanisms at strengths 0.02, 0.05 and 0.10: feature sign flip, feature-wise mean shift, scaling drift, feature dropout with median imputation, random evaluation-label flip and prior-preserving evaluation-label flip. Feature-only sensors are feature Jensen–Shannon divergence (JSD), maximum mean discrepancy (MMD) and the Kolmogorov–Smirnov (KS) rejection rate against the clean version of the same item set; prediction, label-marginal and joint-outcome evidence likewise compare aggregate score, class and confusion-profile distributions with their clean same-item-set values. None uses item pairing. The runner protects these clean arrays as an experimental oracle; the policy API receives only sensor and family flags, but producing those flags still requires the upstream oracle. This is statistical class A, not evidence that a deployed system authenticates the reference. Under $\mathcal{I}_{XFY}^*$, confusion-profile deltas against a trusted aggregate are exact class B invariants, while prediction disagreement and label mismatch use trusted item alignment (class C). The complete sensor-to-reference audit is in the supplement.

The duplicate, clustered and near-constant structure on which this benchmark fingerprint depends is consistent with documented CICIDS2017 data-quality limitations [60]; the frozen staging is described rather than re-engineered here.

5.2 Experimental Units and Uncertainty

Gate 1 crosses five split seeds with four nested model seeds; its 7,980 raw rows contain 3,420 exact repeated SVC rows, leaving 4,560 unique observations. The expansion crosses the same five split seeds with two model seeds; all 360 job pairs completed and 13,680 raw rows contain 2,280 verified repeats, leaving 11,400 unique observations. Coverage endpoints are exact counts in the prespecified design; for the secondary model-profile endpoint (supplement) the five split seeds are the inferential units with two-sided 95% Student- t intervals (four degrees of freedom), describing within-environment resampling only.

5.3 Null Calibration and Decision-Level Calibration

Gate N (preregistered in artifact 1.1.0) calibrates the ten distributional sensors per (environment, dimension, model) cell at a nominal per-sensor $\alpha = 0.05$: the threshold is the

191st of 200 clean calibration draws from one half of the held-out evaluation pool, and false alarms are measured on 200 draws from the disjoint other half. Gate F calibrates each regime as a family with Proposition 5(b)’s conformal rule at $\alpha = 0.05$ per decision, using only the calibration draws. The same draws also evaluate the union and asymmetric comparison rules (Proposition 5(c)). The asymmetric rule’s pooled excess over nominal is decomposed into rule bias (its rate minus the conformal rate on identical draws) and design effect (the conformal rate minus nominal). Both rules are also evaluated under 30 exchangeable re-splits of the 400 pooled draws per cell. The final diagnostic values had been observed before amendment A2 was recorded; A2 disclosed that fact before frozen-only regeneration. Because a run’s 20 draws overlap within one pool half, false-alarm inference uses (environment, split-seed) clusters. Pooled rates are descriptive; intervals are five-cluster t intervals per environment. E1 (256-row draws from 322-row halves) remains in the preregistered primary aggregate and is also reported separately.

5.4 Decision Layer and Offline End-to-End Evaluation

Gate D composes sensors, calibrated rules, the information regime, the trusted-reference status, a materiality threshold and the declared residual blind region into one `allow/hold/block` decision. `block` is reserved for violations of an exact invariant against a trusted reference; `hold` answers statistical evidence or, under P3, missing coverage; the decision composes with the four frozen contracts by the maximum in `allow < hold < block`. Four policies are fixed and named by what they are: P0 serve-always, the baseline; P1, the uncalibrated union rule of Gate N (risk-tolerant); P2, the conformal family rule, a calibrated *risk-tolerant* policy that may serve under an explicitly declared residual blind region; P3, the *coverage-complete abstaining* variant, which holds whenever a mandatory protected boundary (feature, prediction, label) has neither exact nor declared statistical coverage under the regime. P3 fails closed on missing coverage, not on insufficient power: a boundary covered by a low-power calibrated sensor is served as under P2, so P3 guarantees no minimum detection power. The prototype’s contracts fail closed on their invariants; P2 does not and is never called fail-closed. Five regimes are evaluated: $\mathcal{I}_X, \mathcal{I}_{XF}, \mathcal{I}_{Y_m}, \mathcal{I}_{XFY}$ (class-A statistical aggregate comparisons) and $\mathcal{I}_{XFY}^*$ (class-B aggregate and class-C item-aligned exact invariants; feature coverage remains statistical). The policy consumes union/family/exact flags, not clean arrays; the upstream scorer has the reference dependency described above. The observation set is frozen: 12,000 clean evaluation draws, 1,200 near-null synthetic controls applied in place (Gaussian noise $\sigma = 0.001$, scaling $\alpha = 0.001$; not operational benign traffic), and the 10,800 intervened observations of the eight environments, 7,008 of which change the reported balanced accuracy (structural-sensitivity endpoint; $\tau = 0.02$ and 0.05 are larger-effect sensitivities, not operational risk thresholds). The primary endpoint is a materially altered audit result served: a material observation decided `allow`. The CSV retains `unsafe_allow` as a compatibility identifier; *unsafe* here means serving a materially corrupted audit/report conclusion, not allowing a malicious

network event and not necessarily changing the classifier’s live prediction. The cost endpoints are the clean false-action rate (clean draws not allowed; for $\mathcal{I}_{XFY}^*$ the denominator is the 1,200 exact-zero rows) and the interruption rate on the near-null controls (the same 1,200 rows for every regime), decomposed into statistical holds and exact-reference blocks. The evaluation is offline, on frozen outputs, not a deployed runtime service; twelve consistency checks fail closed, and every table and figure is generated from the manifested outputs.

5.5 Adversarial Gate: Cluster-Preserving Perturbation

Gate A executes the preregistered cluster-preserving attacker. The standardized projected features hold tight clusters of rows (in every environment some feature keeps 69–87% of the rows within 0.01 standard deviations) that a fresh clean batch reproduces and any in-place displacement smears, which is why mean shift and scaling drift were detected in every cell. The attacker holds the batch, knows the sensor definitions and the fingerprint, and applies the executed mechanisms unchanged to every unclustered entry (an entry is clustered when at least $\lceil 0.05n \rceil$ rows, never fewer than two, lie within 0.01 batch standard deviations of it); clustered entries and labels are untouched. Strengths 0.02, 0.05, 0.10 match the frozen suite and 0.25, 0.50 probe whether materiality can be bought on the unclustered entries; the executed mechanisms run in the same jobs as matched controls and must reproduce the frozen expansion exactly. The gate runs in the eight environments with the frozen maps, caps and seeds (240 exact-statevector jobs) and is scored with the calibration draws, rules and policy layer of Gates F and D; window, minimum mass and strength grid were fixed before execution.

5.6 Quantum Gate and Executable Prototype

A separate simulator gate isolates the quantum boundary with CICIDS, 64/64 samples, five split seeds, dimensions 4, 6 and 8 and 11 conditions (165 cells): clean, benign level-1 transpilation and a common post-feature-map rewrite as controls; data-dependent RZ mutations at 0.02 and 0.10; a feature-map repetition change; asymmetric, diagonal-eroded and PSD-preserving kernel edits; and binomial fidelity-estimation emulators at 256 and 1,024 shots. Each cell records canonical OpenQASM 3 provenance, kernel hashes, the anchored comparison against the trusted reference kernel, algebraic checks, repeated-estimation discrepancy and prediction changes. The prototype instantiates four ordered contracts (input/preprocessing, circuit/kernel, execution/result, evaluation/report) in an SHA-256 chain over one compact CICIDS cell and six prespecified scenarios; it evaluates composition and decision logic, not throughput. It is the Paper 1.5 research instance of the ATHENA-AEGIS HSaaS component, not evidence of deployment, provider attestation, confidentiality, QPU protection or Fleet Management.

6 RESULTS

6.1 Exact Label-Path Blind Regions

The zero-response cells are validation checks of the structural prediction, not findings: Corollary 1 says that a sensor

whose input does not change cannot respond, and the checks confirm that the implementation realises the declared views. All 3,600 expansion and 1,440 Gate-1 evaluation-label observations (of 11,400 and 4,560 unique observations) have zero prediction disagreement and exactly unchanged feature JSD, MMD, KS rejection and score JSD; all 1,800 and 720 prior-preserving observations also have zero prior shift and label JSD; the verifier recomputes every count. The informative quantity is how often these blind regions coincide with material conclusion changes and, in Section VI-C, with serving decisions. The signed conclusion change is not one-sided: in the expansion 2,184 label observations lower the reported balanced accuracy, 983 leave it unchanged and 433 raise it (Gate 1: 1,276, 105, 59). Every one of the 2,617 (Gate 1: 1,335) label observations with a changed conclusion has non-zero item-aligned confusion evidence, and every one of the 2,184 with a lowered conclusion, the count used by the one-sided diagnostic, is among them.

6.2 Decision-Level Calibration

Over the 12,000 evaluation draws the per-sensor false-alarm rates lie between 0.029 and 0.069, but the batch-level rule “fire if any sensor fires” is not calibrated at the level of the decision: its rates are 0.125, 0.203, 0.048 and 0.259 for $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$ and $\mathcal{I}_{XFY}$ (three, six, two perfectly dependent and ten sensors). The conformal family rule brings them to 0.056, 0.058, 0.048 and 0.053 (Fig. 1(a)), observed rates in a design that violates the exchangeability premise. The asymmetric comparison rule gave 0.061, 0.073, 0.048 and 0.079 on the same draws and attributed its whole excess over 0.05 to the design. That attribution was wrong: the rule bias is +0.005, +0.015, +0.000 and +0.025, the design effect +0.006, +0.008, -0.003 and +0.003, and under exchangeable re-splits of the pooled draws, where any excess is the rule’s own, the earlier rule fires at 0.052, 0.058, 0.040 and 0.059 while the conformal rule stays at 0.044, 0.042, 0.040 and 0.036 against its level 0.0498. What remains with the conformal rule is the design effect, concentrated in E1, whose 200 draws per cell carry the information of about one independent batch per pool half (0.125, 0.115, 0.123 in $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{XFY}$); without E1 the conformal rule sits at 0.048, 0.052, 0.048 and 0.045; E1 stays in the primary aggregate. A split construction (0.049, 0.039, 0.067, 0.034; supplement) is not adopted because it splits the draws between ranking and calibration, reducing the calibration sample and coarsening p -value resolution to 1/101.

As a consistency check of Proposition 4(i), the calibrated rules confirm the structural prediction: on all 3,600 label observations the per-sensor and the conformal $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ rules, and on all 1,800 prior-preserving observations the $\mathcal{I}_{Y_m}$ rules, fire zero times, as they must when their input is unchanged. The informative result is on the other side: the batch-level $\mathcal{I}_{XFY}$ rules detect 43 (union) and 11 (conformal) of the 2,617 material label observations, 0.016 and 0.004, although every one of them is separable (Proposition 3, C8): a 2–10% relabeling of a 128-row batch shifts the confusion profile by less than its variability across fresh clean batches. A trusted aggregate reference of the same batch detects all 2,617 exactly (Corollary 3a) and 2,792 of the 3,600 label rows in total; the 808 relabelings that preserve every

aggregate (C1) are exposed only by the item-aligned label reference (Corollary 3b). Feature-side coverage is mechanism-dependent: mean shift and scaling drift are detected in 0.99–1.00 of $\mathcal{I}_X$ cells, 0.96–1.00 of $\mathcal{I}_{XF}$ cells and 0.95–1.00 of $\mathcal{I}_{XFY}$ cells because the standardized projected features hold tight clusters that an in-place perturbation smears and a fresh batch reproduces, a legitimate but fragile fingerprint that also fires on the near-null controls (40–64% with the conformal rule); feature dropout with imputation is nearly invisible at the batch level (0–3%) yet changes predictions in 49–76% of cells. Calibration costs little detection: containment in $\mathcal{I}_{XF}$ is 0.40 under the union and 0.38 under the conformal rule at the primary threshold, 0.70 versus 0.66 at $\tau = 0.05$.

6.3 Offline End-to-End Decisions

Table 2 and Fig. 1(b,c) report the offline end-to-end evaluation. All aggregate served fractions summarize the prespecified equal-weight intervention grid; they are not deployment prevalence or expected operational unsafe rates. Under P0 every one of the 7,008 material observations is served. Under the batch-level regimes the conformal policy P2 serves 4,496 ($\mathcal{I}_X$), 4,365 ($\mathcal{I}_{XF}$) and 4,390 ($\mathcal{I}_{XFY}$) of them with clean false-action rates 0.056, 0.058 and 0.053, whereas the uncalibrated union serves 4,429, 4,231 and 4,180 at 0.125, 0.203 and 0.259: the union buys 0.40 versus 0.37 containment in $\mathcal{I}_{XFY}$ at more than four times the clean false-action rate and 0.76 versus 0.49 interruption of the near-null controls. The asymmetric comparison rule served 4,494, 4,327 and 4,322; the adopted rule moves the count by less than 1% of the material rows. The served material corruptions are dominated by structure, not by thresholds: all 2,617 material label observations are served under $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ (residual blind region), 2,606 of them under batch-level $\mathcal{I}_{XFY}$ (statistically undetectable), and 1,057 of 1,079 material dropout observations under $\mathcal{I}_{XF}$. The $\mathcal{I}_{Y_m}$ view provides zero containment of material interventions in this suite; 5,450 material cases remain in its residual blind region at $\tau \rightarrow 0^+$, while all 7,008 material cases are served once both blind and statistically missed cases are counted.

Containment 1.00 follows constructively from Corollary 3 under the declared trusted references; the experiment quantifies its operational/statistical cost. Consistently, the trusted-reference regime serves none of the 7,008 material observations with zero clean false actions on its 1,200 exact-zero rows and blocks every non-identity relabeling and every prediction change through its item-aligned references. Its cost has two parts. Of the 1,200 near-null synthetic controls the trusted calibrated policy interrupts 629: 544 are statistical holds by its batch component, and 85 are gross exact-reference blocks of controls whose predictions changed under a 0.001 perturbation. Of those exact blocks, 46 already belong to the 590 interruptions produced by batch $\mathcal{I}_{XFY}/P2$. The net increment is therefore 39/1,200, or 3.25 percentage points, not the gross 85/1,200. The controls are synthetic near-null variation, not operational traffic, so neither number is a production false-positive rate, and the two denominators of Table 2 are reported side by side. The coverage-complete abstaining policy P3 makes information-set conditionality literal: in $\mathcal{I}_X$, $\mathcal{I}_{XF}$ and $\mathcal{I}_{Y_m}$ the label boundary has no coverage, so nothing is served and every clean and near-null row is interrupted; in $\mathcal{I}_{XFY}$ and $\mathcal{I}_{XFY}^*$ every boundary

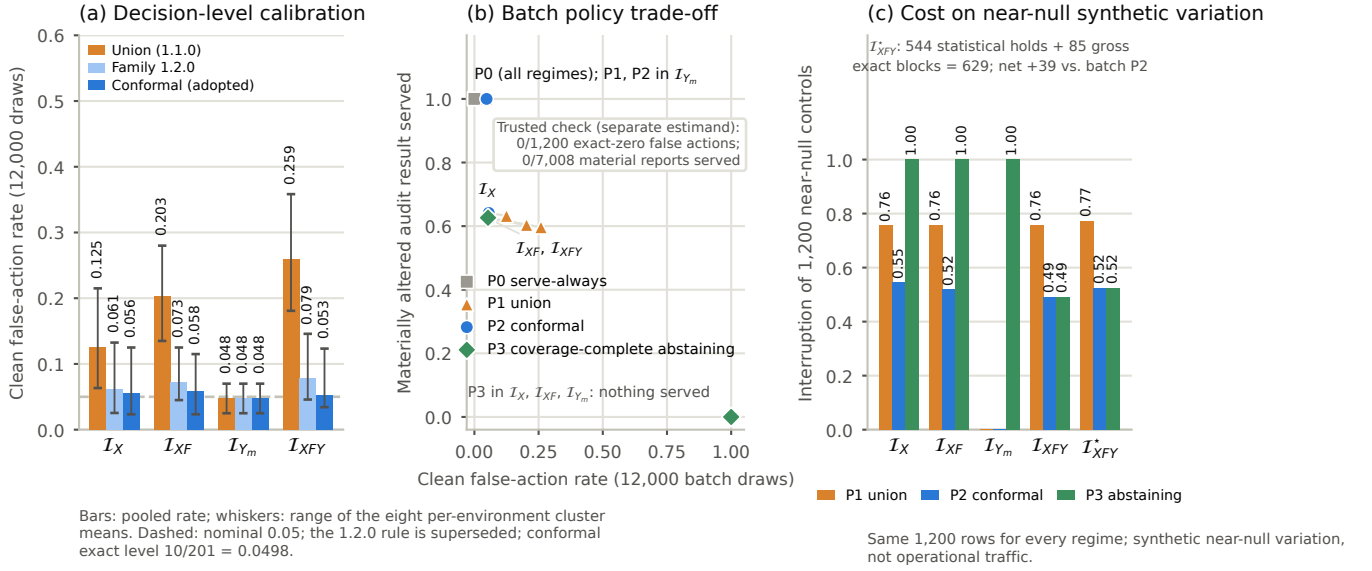


Fig. 1. Gates F and D (frozen evidence of artifact 1.3.0). (a) Decision-level false-alarm rate on 12,000 disjoint clean draws of the union rule, of the asymmetric comparison rule and of the conformal family rule per regime; whiskers span the eight per-environment cluster means. (b) Rate of materially altered audit results served against clean false-action rate for each batch policy and regime at the structural-sensitivity endpoint; the batch FPR denominator is 12,000 draws. The trusted invariant check is reported separately because its denominator is 1,200 exact-zero rows and it is a different estimand. The coverage-complete abstaining policy P3 serves nothing where the label boundary has no coverage. (c) Interruption of the same 1,200 near-null synthetic controls by each policy and regime; the trusted regime’s origin in (b) costs 629 controls here, 544 statistical holds and 85 gross exact-reference blocks.

TABLE 2

Offline decisions on frozen observations at the structural-sensitivity endpoint $|\Delta_R| > 0$ (7,008 materially altered audit results among 10,800 interventions). P1: union; P2: conformal, risk-tolerant; P3: coverage-complete abstaining. FPR is estimated on 12,000 disjoint clean draws for batch regimes; the starred row instead reports an exact-zero invariant check on 1,200 rows and is not the same estimand. Benign is interruption of the same 1,200 near-null synthetic controls. Served: undetected materially altered audit results served, not malicious network events. Blind: structurally indistinguishable material rows.

Regime	Policy	FPR/check	Benign	Served	Contain.	Blind
I_X	P1 union	0.125	0.76	4,429	0.37	2,617
I_X	P2 conf.	0.056	0.55	4,496	0.36	2,617
I_X	P3 abst.	1.000	1.00	0	1.00	2,617
I_{XF}	P1 union	0.203	0.76	4,231	0.40	2,617
I_{XF}	P2 conf.	0.058	0.52	4,365	0.38	2,617
I_{XF}	P3 abst.	1.000	1.00	0	1.00	2,617
I_{Y_m}	P1 union	0.048	0.00	7,008	0.00	5,450
I_{Y_m}	P2 conf.	0.048	0.00	7,008	0.00	5,450
I_{Y_m}	P3 abst.	1.000	1.00	0	1.00	5,450
I_{XFY}	P1 union	0.259	0.76	4,180	0.40	0
I_{XFY}	P2 conf.	0.053	0.49	4,390	0.37	0
I_{XFY}^*	P2/P3 (exact)	0.000	0.52	0	1.00	0

is covered and P3 coincides with P2, including P2’s 4,390 materially altered audit results served in I_{XFY} : abstention on missing coverage is not a guarantee of power, and the choice between P2 and P3 is a declared risk decision, not a detector property. The composition with the six frozen contract envelopes reproduces their actions (supplement).

6.4 Adaptive Cluster-Preserving Attacker

Gate A ran as preregistered: 240/240 jobs, with exact replay of clean rows and matched controls. This is a stress test of a fragile cluster-dependent fingerprint, not the defeat

of a generally powerful drift detector. At strengths 0.02–0.10, detection in the feature and feature-plus-prediction regimes falls from 0.96–1.00 to 0.01–0.66 while retaining 83–91% of materiality. Among 3,418 material adaptive rows, P2 serves 1,328/959/ 1,006 in $I_X/I_{XF}/I_{XFY}$, aggregate rates 0.39/0.28/ 0.29; I_{XFY}^* blocks every material row. The rate is strength-dependent. In I_{XFY} , mean-shift material served rates at 0.02/0.05/0.10/0.25/0.50 are 0.90/0.67/ 0.46/0.16/ 0.08, and scaling rates are 0.55/0.36/ 0.22/0.16/ 0.10. The supplement reports materiality and detection for every strength and regime. The attack is most effective while subtle; stronger perturbations increasingly shift the aggregate and are easier to detect.

The high matched-control response, near-null sensitivity and adaptive evasion are three manifestations of the same duplicate/cluster structure, not three independent validations. The result is a property of the fingerprint, not of the mechanism: calibration controls false alarms but buys no coverage against an attacker who knows what the sensor sees; only a reference does.

6.5 Quantum-Workflow Coverage

All 165 cells are present and all nine prespecified acceptance checks pass; the supplement tabulates and plots the response fraction of every sensor per condition, read class by class. In this ideal-statevector gate the honest estimate equals the semantic kernel and g is the identity, so one anchored comparison against $\Phi(C_0)$ serves as the semantic probe on circuit-side rows and as the observed-kernel reference on post-processing rows; under finite-shot estimation the two anchors separate (Proposition 7(iii)). Clean cells trigger no sensor. Benign transpilation and the common-unitary rewrite

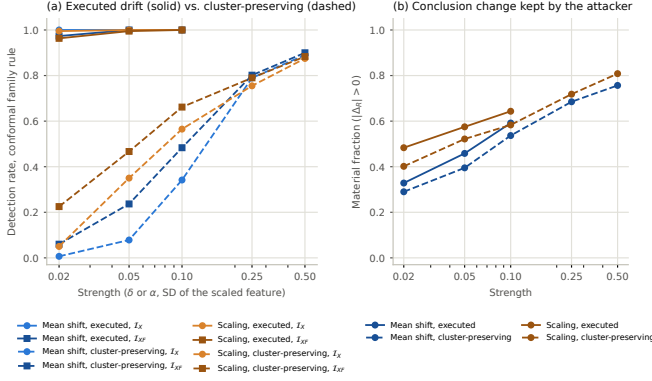


Fig. 2. Gate A. (a) Detection under the conformal family rule of the executed drift mechanisms (solid) and of their cluster-preserving variants (dashed) in $\mathcal{I}_X$ and $\mathcal{I}_{XF}$, 600 observations per point. (b) Material fraction of the same rows: the attacker evades the batch-level sensors at matched strengths while keeping most of the conclusion changes.

change serialized provenance while preserving semantic kernels, algebraic properties and outputs in every cell (Proposition 7(i)): a raw hash difference needs policy-aware adjudication. Every data-dependent circuit mutation changes the semantic kernel; the mild RZ mutation changes no prediction, the stronger one changes predictions in 2/15 cells and the repetition change in 11/15. Asymmetric and diagonal-eroded matrices are detected algebraically in every cell and change no prediction; the PSD-preserving mixture is symmetric, unit-diagonal and PSD in all cells, so algebraic validation is exactly blind, the circuit hash sees nothing because C is unchanged, and only the kernel hash and the anchored comparison of the observed kernel expose all 15 substitutions, with output monitoring reacting in 1/15 (Proposition 7(ii)). Both shot emulators produce non-zero repeated-estimation discrepancy in all 30 cells and output changes in 7/15 and 3/15, a property of these cells rather than the universal law that Proposition 7(iii) declines to state; the contract holds them for adjudication instead of testing exact equality. The prototype’s six envelopes verify and its eight acceptance checks pass (clean and approved transpilation allowed; parameterized mutation and PSD-preserving substitution blocked; prior-preserving label corruption blocked at the evaluation contract despite a zero class-prior delta; the approved 256-shot emulator held). The secondary ZZ-versus-SVC comparison (supplement) is retained only to show that the primary result does not depend on the model: its direction is stable, its magnitude conditional on scalars and clean headroom, and it is neither a quantum-advantage nor a vulnerability ordering.

7 INTEGRITY-AUDIT CONTRACT

The resulting contract requires each claim to name its boundary, protected asset, intervention class, fixed elements, view, reference profile, sensor, calibration premise, residual blind region, near-null cost and response. For labels, a trusted aggregate plus joint audit suffices for conclusion integrity; item identity requires item-level provenance. For the studied quantum path, the bundle combines circuit/parameter provenance, approved-transpilation policy, semantic probes,

observed-kernel reference, algebraic checks and repeated-estimation evidence. Violated exact invariants block; missing mandatory evidence holds. P2 may serve within a declared residual region; P3 abstains on missing coverage but promises no minimum power.

8 DISCUSSION AND LIMITATIONS

8.1 Security Interpretation

A metric can change, or even improve, while predictor output is bit-for-bit unchanged. Calling this model degradation assigns failure to the wrong component: input robustness cannot repair corrupted ground truth, whereas provenance and joint validation can. Zero sensor response is informative only when its evidence could distinguish the states; separability yields practical detection only with a sufficiently tight null. On the prespecified equal-weight stress grid, the macro-reported served fractions are mechanism summaries, not deployment prevalence. The adversarial gate adds that calibration controls false alarms, not what an attacker can hide: after the declared fingerprint is preserved, 28–39% of material instances are served in aggregate, with the full strength-specific profile reported above.

The contribution is not storing and comparing a reference, but deriving its claim-relative minimum granularity. A trusted same-batch aggregate certifies the reported conclusion; item identity needs item alignment. Conclusion integrity need not duplicate the item-wise label store, but every guarantee requires an external root outside the adversary’s rewrite class (Proposition 6). We identify that root’s minimum granularity rather than claim to eliminate it.

The trusted cost is likewise claim-relative. Its 85 exact-reference blocks are a gross mechanism count, not the incremental interruption cost. Because 46 overlap the batch $\mathcal{I}_{XFY}$ /P2 interruptions, the trusted policy adds 39/1,200 (3.25 percentage points) while producing 629 total near-null interruptions. No such quantity is an operational traffic false-positive rate.

8.2 Validity Boundaries

The three datasets and eight fixed scale/temporal settings are not sampled deployments; six of nine environments use CICIDS2017, five split clusters measure within-environment resampling only, most gates use 128/128 samples, and numeric-only balanced staging does not reproduce operational prevalence. Both quantum evaluators are ideal state-vector simulators with finite-shot emulation, not hardware evidence. The semantic-versus-observed anchor distinction is formal; empirical separation under hardware noise belongs to the companion operational study. SVC and QSVC retain pipeline-specific scalars, so their comparison does not isolate kernel geometry; those scalars also govern how far the attacker’s unclustered entries move. Exact blindness is a capability result under constructed perturbations, not a prevalence estimate.

The conformal level is marginal and requires exchangeability of calibration draws and the audited batch. The executed design violates that premise because draws overlap within a pool half, calibration and evaluation use different rows of one table, and E1’s pool half supplies one disjoint

batch. We report the residual excess as executed and claim no conditional guarantee given a fixed calibration set. Naive drift response, near-null interruption and adaptive F5 evasion reflect the same local duplicate/cluster structure, not independent validations. No study environment removes that structure while preserving the rest of the design. Gate A fixes one cluster rule and two mechanisms; other attacker models are unevaluated.

The trusted item-aligned regime is an assumption, not a stronger detector. If the reference can be rewritten with the asset, Proposition 6 applies; its cost is measured on near-null controls, not operational traffic. P3 abstains only on missing coverage, so a nominally covered low-power sensor remains a limitation, and calibration-validity abstention is outside this layer. The local prototype lacks authenticated provider records; its hash chain establishes consistency, not authenticity. Every policy result is offline on frozen outputs. Context-conditioned runtime calibration under non-stationarity, out-of-support abstention with recovery, real-QPU execution and the service-level evaluation of enforcement are the subject of a separate study.

9 CONCLUSION

Hybrid workflow integrity is evidence-relative, not a robustness number. Interventions may be structurally invisible, statistically missed, or made evasive by an attacker who preserves the monitored fingerprint. Across the frozen design, our view model, proofs and calibrated offline policy separate these cases and their near-null cost; the conformal guarantee remains conditional on exchangeability and the quantum validation simulator-bounded. The constructive result is claim-relative minimum evidence: an authenticated same-batch aggregate can certify conclusion integrity, whereas item-identity integrity requires alignment. Assurance must therefore declare its boundary, view, trust root, calibration premise, residual region, cost and response.

DATA, CODE, AND ETHICS STATEMENT

The review artifact contains code, locked dependencies, tests, derived tables, figures, eight evidence manifests, gate preregistrations, amendments and replay commands. Raw CICIDS2017, UNSW-NB15 and ToN-IoT data are not redistributed; sources, hashes and staging are documented. Experimental evidence is frozen at artifact 1.3.0 (version DOI 10.5281/zenodo.22648573). Version 1.3.4 is archived at Zenodo under the concept DOI 10.5281/zenodo.22550852 (version DOI in the artifact metadata); code is released under Apache-2.0, derived evidence, figures and documentation under CC BY 4.0. The study uses previously released benchmarks and involves no new human-participant or personal-data collection.

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corrections, including the reference-by-reference primary-source audit. The authors specified the protocols, verified the primary sources, proofs, numerical results, code and generated text, and retain full responsibility; the submission package contains the full system-by-section disclosure.

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